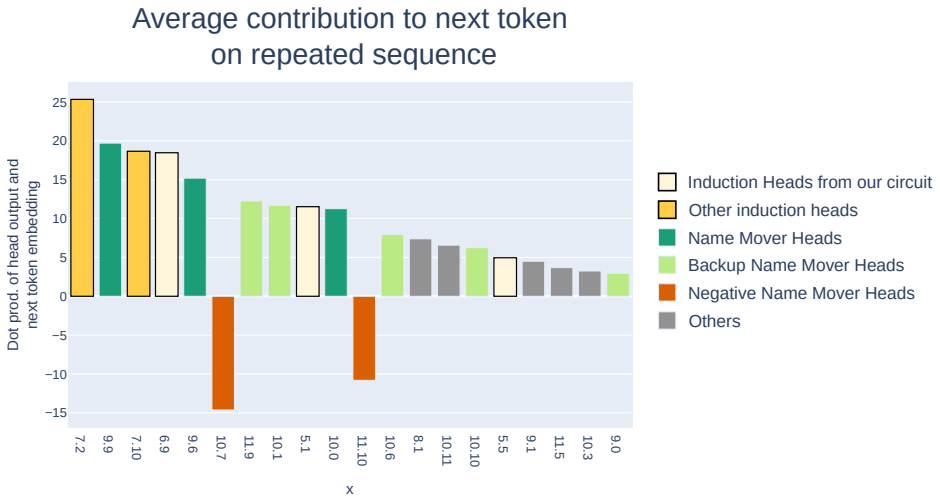

Figure 17: Contribution to the next token prediction per head on repeated sequences of tokens. The heads are ordered by decreasing absolute values of contribution. Black contour: heads with attention patterns demonstrating prefix matching property.

- The *previous token score*: we averaged the attention probability on the off-diagonal. This is the average attention from the token at position i to position $i-1$.
- The *induction score*: the average attention probability from T_i to the token that comes after the first occurrence of T_i (i.e. T_{i-99})

These two scores are depicted in Figure 18 (center and right) for all attention heads.

Previous Token Heads. 4.11 and 2.2 are the two heads with the highest previous token score on sequences of random tokens. This is a strong validation of their role outside p_{IOI}.

Induction Heads. Olsson et al. (2022) define an Induction Head according to its behavior on repeated sequences of random tokens. The attention head must demonstrate two properties. i) Prefix-matching property. The head attends to `[B]` from the last `[A]` on pattern like `[A] [B] . . . [A]` ii) Copy property. The head contribute positively to the logit of `[B]` on the pattern `[A][B]...[A]`.

5.5 and 6.9 are among the 5 heads with the highest induction score. This validates their prefix-matching property introduced in Olsson et al. (2022).

To check their copy property, we computed the dot product $\langle h_i(X), W_U[B] \rangle$ between the output of the head h_i on sequence X and the embedding of the token `[B]` on repeated sequences of random tokens. The results are shown in Figure 17. The two Induction Heads (5.5 and 6.9) appear in the 20 heads contributing the most to the next token prediction. Thus validating their copying property.

We also noticed that the majority of the Negative, Backup and regular Name Mover Heads appear to write in the next token direction on repeated sequences of random tokens, and Negative Name Movers Heads contribute negatively. This suggests that these heads are involved beyond the IOI task to produce next-token prediction relying on contextual information.

I Validation of Duplicate Token Heads

On the repeated sequences of random tokens (Appendix H) we also computed the *duplicate score*. For each token T_i in the second half of a sequence, we average the attention probability from T_i to its previous occurrence in the first half of the sequence (i.e. T_{i-100}).

The duplicate token scores for all attention heads are depicted in Figure 18. 3.0 and 0.1 are among the three heads with the highest duplicate token score (Figure 18). This is evidence of their role of Duplicate Token Heads outside the circuit for the IOI task.

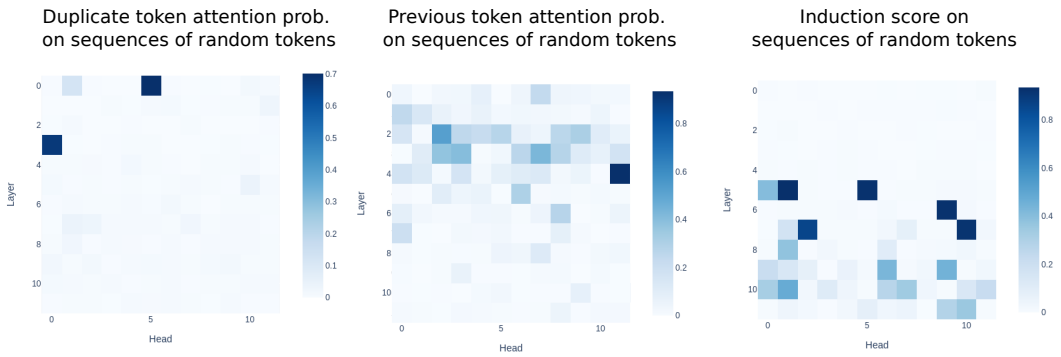

Figure 18: Attention scores on sequences of repeated random tokens. Left: Duplicate score, the average attention probability from a token to its previous occurrence. Center: Previous token attention score, it is the average of the off-diagonal attention probability. Right: Induction score. Average attention probability from the second occurrence of `[A]` to `[B]` on patterns `[A][B]...[A]`.

However, the fuzzy Duplicate Head 0.10 doesn't appear on this test. By qualitatively investigating its attention patterns on Open Web Text, we found that this head attends strongly to the current token. Moreover, when the current token is a name, and it is duplicated, the head attends to its previous occurrence.

J Role of MLPs in the task

In the main text, all of the circuit components are attention heads. Attention heads are the only modules in transformers that move information across token positions – a crucial component of the IOI task – so they were our main subject of interest. However, MLPs can still play a significant role in transforming the information in each residual stream. We explored this possibility by measuring the direct and indirect effects of each of the MLPs in Figure 19. In these experiments, for each MLP in turn, we did a path patching experiment (Section 3.2) to measure the direct effect and a knock-out experiment for the indirect effect.

We observe that MLP0 has a significant influence on logit difference after knock-out (it reverses the sign of the logit difference) but the other layers don't seem to play a big role when individually knocked out. When all MLP layers other than the first layer are knocked out, however, the logit difference becomes -1.1 (a similar effect to the knockout of MLP0 alone).

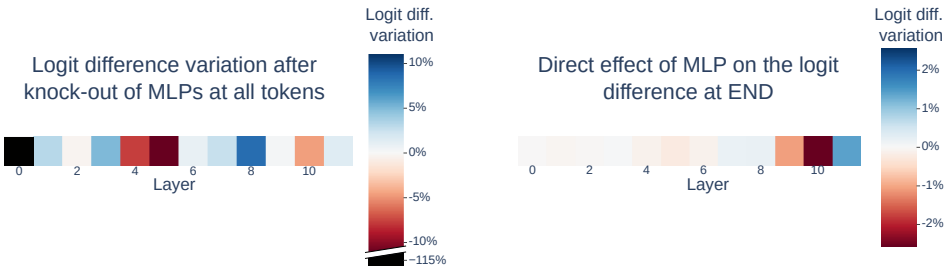

Figure 19: Left: change in logit difference from knocking out each MLP layer. Right: change in logit difference after a path patching experiment investigating the direct effect of MLPs on logits.

K Minimality sets

The sets that were found for the minimality tests are listed in Figure 20.

v	Class	$K \cup \{v\}$	$F(C \setminus (K \cup \{v\}))$	$F(C \setminus K)$
(9, 9)	Name Mover	[(9, 9)]	2.26	2.62
(10, 0)	Name Mover	[(9, 9), (10, 0)]	1.91	2.26
(9, 6)	Name Mover	[(9, 9), (10, 0), (9, 6)]	2.11	1.91
(10, 7)	Negative	[(11, 10), (10, 7)]	4.07	3.13
(11, 10)	Negative	[(11, 10), (10, 7)]	4.07	3.27
(8, 10)	S Inhibition	[(7, 9), (8, 10), (8, 6), (7, 3)]	0.24	1.01
(7, 9)	S Inhibition	[(7, 9), (8, 10), (8, 6), (7, 3)]	0.24	0.92
(8, 6)	S Inhibition	[(7, 9), (8, 10), (8, 6), (7, 3)]	0.24	0.86
(7, 3)	S Inhibition	[(7, 9), (8, 10), (8, 6), (7, 3)]	0.24	0.43
(5, 5)	Induction	[(5, 9), (5, 5), (6, 9), (5, 8)]	0.97	2.12
(5, 9)	Induction	[(11, 10), (10, 7), (5, 9)]	3.33	4.07
(6, 9)	Induction	[(5, 9), (5, 5), (6, 9), (5, 8)]	0.97	1.46
(5, 8)	Induction	[(11, 10), (10, 7), (5, 8)]	3.83	4.07
(0, 1)	Duplicate Token	[(0, 1), (0, 10), (3, 0)]	0.60	1.90
(0, 10)	Duplicate Token	[(0, 1), (0, 10), (3, 0)]	0.60	1.66
(3, 0)	Duplicate Token	[(0, 1), (0, 10), (3, 0)]	0.60	1.05
(4, 11)	Previous Token	[(4, 11), (2, 2)]	1.31	2.28
(2, 2)	Previous Token	[(4, 11), (2, 2)]	1.31	1.75
(11, 2)	Backup Name Mover	All previous NMs and backup NMs	0.95	1.37
(10, 6)	Backup Name Mover	All previous NMs and backup NMs	1.65	1.88
(10, 10)	Backup Name Mover	All previous NMs and backup NMs	1.88	2.11
(10, 2)	Backup Name Mover	All previous NMs and backup NMs	1.49	1.65
(9, 7)	Backup Name Mover	All previous NMs and backup NMs	0.81	0.95
(10, 1)	Backup Name Mover	All previous NMs and backup NMs	1.37	1.49
(11, 9)	Backup Name Mover	All name movers and negative heads	0.41	0.45
(9, 0)	Backup Name Mover	All name movers and negative heads	0.41	0.45

Figure 20: K sets for minimality for each v.